Name: Class: Date:

Activity 1 · Hands-on · 🌳 Outdoors · ⏱ 1–2 hours

📝 Campus Biodiversity Survey

You will learn: Species richness differs between habitats; disturbed places usually have fewer species.

YOU NEED

- Notebook
- Phone camera
- A plant/bird identification app or book (optional)

⚠️ **Safety:** Do not disturb nests or pluck plants. Work in pairs.

STEPS

1. Choose 3 areas: a mowed lawn, a garden with many plants, and an undisturbed corner.
2. In each area, spend 15 minutes listing every different plant, insect and bird you see.
3. Count the number of different species (species richness).
4. Note the most common species in each area.
5. Compare the areas and suggest reasons for the differences.

OBSERVATION TABLE

Area	Number of species	Most common species	Disturbance (low / medium / high)

QUESTIONS

1. Which area had the highest species richness? Why?

2. What is the evil quartet?

3. Suggest two ways to increase biodiversity on your campus.

WHAT IS HAPPENING? — THE SCIENCE

Habitats with more layers and less disturbance support more species. Mowing, paving and pollution reduce habitat — an example of habitat loss, one of the "evil quartet" causes of biodiversity loss.

Key terms: **species richness** **habitat loss** **evil quartet**

Activity 2 · Hands-on · 🌳 Outdoors · ⏱ 1 hour

📐 Species–Area Plot

You will learn: Species richness increases with area: log S = log C + Z log A (Humboldt).

YOU NEED

- String and sticks to mark squares of 1, 4, 16 and 64 m² (nested)
- Notebook
- Calculator

⚠️ **Safety:** Wear shoes; watch for insects.

STEPS

1. Mark a 1 m × 1 m square and count the different plant species in it.
2. Expand to 2 m × 2 m, then 4 m × 4 m, then 8 m × 8 m, counting the total species each time.
3. Fill the table and calculate log A and log S.
4. Plot S against A, then log S against log A.
5. Draw the best straight line on the log graph and estimate its slope (Z).

OBSERVATION TABLE

Area A (m²)	Species S	log A	log S

QUESTIONS

1. What shape was your S vs A graph?

2. What did you get for Z? How does it compare with 0.1–0.2?

3. Why is protecting large areas important?

WHAT IS HAPPENING? — THE SCIENCE

Bigger areas include more microhabitats, so they hold more species — but each doubling adds fewer new species. On a log–log graph the relationship becomes a straight line whose slope is Z (about 0.1–0.2 for small regions).

Key terms: **species–area relationship** **Humboldt** **Z value**

Digital activity

💻 Species–Area Explorer

> Change C and Z and see how species richness grows with area on normal and log scales.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch13/2**

> **You will learn:** S = C·A^Z. Larger Z means species richness rises faster with area — as seen for whole continents (Z ≈ 0.6–1.2).

THINK & WRITE

1. With Z = 0.15, how many times must the area grow to double the species?

2. Why is Z larger for very large areas like continents?